

M13-C3 — H3 BK-leading distractor

ECAI target c retains isolated A

H3 run / analysed / not a Bucket 3 claim

1 Status

H3 run / analysed. ECAI (primary) and AAMAS (secondary) on fixture `m13_bucket3_binary_bd_and_lead_a / n30_seed42` are complete. Markdown twin: `h3_bk_leading_distractor.md`.

Lead finding: BK-leading isolated a is pulled into ECAI's theory for deterministic child c (first fold selects `a_val_*`; delta conditioned on a ; not the evaluator BD AND). **Secondary:** brave solved can rely on per-row assumption choice where BK underdetermines the target. H0–H2 remain closed. **Not a Bucket 3 claim.** H4 remains deferred / infra-blocked.

2 Research question

On a BD AND collider with BK-leading independent isolated root A , does ECAI learning of target c select a in the first fold and retain a in the final ABA framework rather than returning the evaluator-only rule `c(A) :- b_val_1(A), d_val_1(A).?`

3 Main finding (lead)

On frozen sample `n30_seed42`, ECAI **solves** target c , but BK-leading isolated a is pulled into the learned theory.

Decisive first fold:

```
folding result: c(A) <- [a_val_1(A)]
```

then assumption introduction of α_1 . Both ordinary target rules remain conditioned on a . Emitted delta:

```
c(A) :- alpha_1(A), a_val_1(A).
c(A) :- alpha_2(A), a_val_0(A).
c_alpha_1(A) :- b_val_0(A).
c_alpha_1(A) :- alpha_3(A), b_val_1(A).
c_alpha_2(A) :- d_val_0(A).
c_alpha_2(A) :- b_val_0(A).
c_alpha_3(A) :- alpha_1(A), a_val_1(A).
```

(plus `assumption/contrary` declarations).

Object	Content
Evaluator-only reference for c	$c(A) :- b_val_1(A), d_val_1(A). (\text{no } a)$
ECAI metrics body variables for c	$\{a\}$
ECAI delta	assumption-rich; both target rules involve a
Among $C=1$ on $n=30$	all 15 have $(b, d) = (1, 1)$; both values of a (10/5)

Distractor inclusion / first-fold distraction away from the mechanism-aligned B, D AND. **solved** + audit SAT does **not** mean the evaluator BD rule was recovered.

4 Secondary finding (brave residual)

ECAI brave **solved** can be sample-adequate via **per-row assumption choice** even when BK features do not determine the target. On H3 ECAI c , this is concentrated in the nested $a=1 / \alpha_1\text{--}\alpha_3$ repair. The $a=0 / c_alpha_2$ side is more BK-readable. Clearer prior exemplar: H0 ECAI target a , rows 9 vs 26 (`m13_bucket3_binary_collider_and`).

Bound: **solved** + **audit SAT** $\neq$ **BK $\rightarrow$ target predictor recovered** / **mechanism-aligned**.

5 Fixture / setup

Fixture `m13_bucket3_binary_bd_and_lead_a`: $B \rightarrow C \leftarrow D$; isolated $A \sim \text{Bernoulli}(1/2)$; $B \sim \text{Bernoulli}(4/5)$; $D \sim \text{Bernoulli}(7/10)$; $C = B \wedge D$. For target c , BK predictors are a , then b , then d (leading distractor). Evaluator reference: $c(A) :- b_val_1(A), d_val_1(A)$. Primary arm ECAI; secondary AAMAS; same frozen `n30_seed42`.

6 AAMAS contrast

Same sample. AAMAS solves c with

```
c(A) :- a_val_1(A), b_val_1(A), d_val_1(A).
c(A) :- a_val_0(A), b_val_1(A), d_val_1(A).
```

so AAMAS also retains `a_val_*` (distractor omission failure; different shape from ECAI). Versus H2: trailing isolated d under AAMAS retained `d_val_*`; here leading isolated a is retained under both arms.

7 Secondary targets

ECAI: all of a, b, c, d **solved**. AAMAS: c **solved**; a, b, d **completed_no_solution**. Other targets are infrastructure retention only.

8 Boundaries / non-claims

- Not a Bucket 3 claim.
- Inclusion of a is not causal recovery of an $A \rightarrow C$ edge.

- **solved** / SAT is not automatically mechanism-aligned.
- “ECAI always solves” is not the H3 question.
- H0–H2 unchanged except for cross-links. H4 untouched.

9 Next

H3 documentation is complete. H4 remains proposed / blocked on target-wise cautious support.
Still **no Bucket 3 claim**.